

Lemma 3.1. If $|Z| \geq M(t)$, every 2-colouring of the subsets of Z has a monochromatic t -cube $\{U \cup X_I : I \subseteq [t]\}$ inside Z .



one of $(t+2)^M$ patterns serves many σ

Proposition 3.3. For every $r \leq n - M$ some shape $(u; a_1, \dots, a_t)$ with $r \leq u \leq r + M - t$ satisfies $\rho_X(u; a) \geq (t+2)^{-M}$.



tune $t+1$ block sizes, fix the rest

Lemma 4.1. For each level j and generator set A , at least M^{m-t-1} of the M^m size vectors β give the cube $Q(j, A)$ a good shape.



average over (σ, β)

Lemma 4.2. Some ordering σ and sizes β make at least $|\mathcal{P}|/((t+2)^M M^{t+1})$ of the cubes monochromatic at once.



union rule, Lemma 2.4

Proposition 2.5. Monochromatic cubes of one block chain glue into a monochromatic union-closed family, so $F(n) \geq |\mathcal{P}|/(2(t+2)^M M^{t+1}) \geq c_t n^{t+1}$.